BUREAU OF IMAGINARY SOLUTIONS	DOC BIS-046 · REV Ω	READING ROOM

THE BUREAU READER · BOOK TWO

THE DOCKET — 45 HYPERSTITION FILINGS, BY SYZYGY

OWN-VOICE CORPUS — FICTION / GAME ASSET

> ***A fiction, filed to production spec.*** *Part of THE BUREAU READER (Doc BIS-046) — the complete training corpus of CLINAMEN-45B-A9B, reproduced verbatim, own-voice only. Nothing here is a prediction, an offer, or advice. Companion at ubu.numetal.xyz/reader.*

Book Two — The Docket

Forty-five hyperstition filings — falsifiable near-future fictions of hard tech and space futures. Each carries a horizon year and a real public source against which, in the game, it will one day read as manifested or rotted. Ordered by syzygy, then by ascending swerve.

SYZYGY 9::0 — ESCHATOLOGY & SILENCE

> *Terminal engineering. Last-photon problems, decommissioning at civilizational scale, deliverables that are absences. The docket opens where things end.*

BIS-F-2026-0001 — high-performance computing

Zone 0 · Horizon 2028 · Swerve 4.318 · Status: pending

FILING 0/086-DELIST. An exascale system commissioned before 2024 will be absent from the TOP500 list by its June 2028 edition — final ranking archived, machine hall returned to the host laboratory's facilities division. The metric of record is negative space: a last appearance in the list's historical database, then nothing at that site for two consecutive editions. Twenty-odd megawatts of demand handed back to the regional grid. The Bureau confirms receipt of the silence.

Resolution source: TOP500 list historical database

BIS-F-2026-0002 — fusion decommissioning

Zone 9 · Horizon 2041 · Swerve 5.641 · Status: pending

FILING 9/333-GRAM. Decommissioning of Europe's largest tokamak — final plasma 2023 — will pass its tritium milestone by 2041: on-site inventory below one gram, confirmed in the operating laboratory's published annual environmental and decommissioning reports. The Bureau notes the asymmetry without comment elsewhere on this form. The machine that demonstrated deuterium-tritium burn will be certified empty while its successors are still certifying concrete. File under terminal engineering, subsection: fuel cycles that close before the next one opens.

Resolution source: UKAEA published annual environmental and decommissioning reports

BIS-F-2026-0003 — subsea optical communications

Zone 9 · Horizon 2039 · Swerve 6.020 · Status: pending

FILING 9/615-ABYSS. By 2039 a transoceanic cable of the first coherent-optics generation will be formally retired and abandoned in place — status amended in the TeleGeography submarine cable database, out-of-service notice circulated per International Cable Protection Committee practice. Recovery will be assessed and declined; the economics of lifting nine thousand kilometers of armored glass do not close. The seabed thereby acquires a permanent stratum of retired bandwidth, and every future route survey must chart around a fossil.

Resolution source: TeleGeography submarine cable database; ICPC out-of-service notices

BIS-F-2026-0004 — semiconductor fabs

Zone 0 · Horizon 2036 · Swerve 6.457 · Status: pending

FILING 0/207-TAPEOUT. A US foundry will close its last 200 mm wafer line by 2036, the terminal lot documented in published process-change notifications and the closure logged in industry fab-capacity records. Legacy nodes do not die of obsolescence; demand for 180 nm parts persists. They die of spare-parts exhaustion — the final refurbished stepper cannibalizing its last donor. The certificate is pre-stamped. Somewhere a 1997 lithography tool is already the only one of its kind still holding vacuum.

Resolution source: Published fab process-change notifications; SEMI World Fab Forecast closure records

BIS-F-2026-0005 — fission decommissioning

Zone 9 · Horizon 2033 · Swerve 7.212 · Status: pending

FILING 9/114-DECOM. By 31 December 2033, at least six pressurized-water reactors of Europe's 1980s build cohort will carry the status PERMANENT SHUTDOWN in the IAEA PRIS registry, cumulative retired capacity exceeding 7 GWe. The Bureau does not predict the shutdowns. It predicts the clerical lag: registry status changes trailing final grid disconnection by a median of eleven weeks, during which each reactor exists in neither column of the ledger. Verification quarterly, against the PRIS status field.

Resolution source: IAEA PRIS reactor registry, status field

BIS-F-2026-0006 — grid energy accounting

Zone 9 · Horizon 2034 · Swerve 7.883 · Status: pending

FILING 9/440-NULL. A G7 transmission system will record its final coal-fired megawatt-hour no later than 2034. The Bureau does not name the unit; it fixes the accounting. Generation-by-fuel-type on the ENTSO-E Transparency Platform will show the coal column reach zero and hold zero for four consecutive quarters — one hundred and fifty years of combustion closed out as a null entry, while the ash ponds remain on the books until 2070.

Resolution source: ENTSO-E Transparency Platform, generation-by-fuel-type dataset

BIS-F-2026-0007 — orbital logistics

Zone 0 · Horizon 2031 · Swerve 8.104 · Status: pending

FILING 0/031-REENTRY. Every spacecraft of a commercial broadband constellation's first design block — more than 3,000 catalog objects — will be confirmed decayed by 2031, checked against the public satellite catalog and decay listings maintained by the 18th Space Defense Squadron. Each object received an entry; each will receive an exit. This is the first infrastructure whose end-of-life plan is combustion over open ocean, executed on schedule, without ceremony, at a rate of roughly four funerals per day.

Resolution source: Space-Track.org satellite catalog and 18th Space Defense Squadron decay listings

BIS-F-2026-0008 — crewed orbital infrastructure

Zone 0 · Horizon 2032 · Swerve 8.907 · Status: pending

FILING 0/500-STATION. The station assembled in low Earth orbit between 1998 and 2011 will complete its targeted final reentry over the South Pacific by 2032, at which point catalog number 25544 transfers from the active satellite catalog to the decayed-objects file at Space-Track.org. Four hundred twenty tonnes. Thirty-plus years of continuous habitation. One debris footprint inside the agreed corridor. The Bureau's interest is confined to the burn itself: the largest deliberate deorbit ever attempted, graded pass/fail, no partial credit.

Resolution source: Space-Track.org catalog entry 25544, decayed-objects file

BIS-F-2026-0009 — deep-space optical/RF communications

Zone 9 · Horizon 2030 · Swerve 9.412 · Status: pending

FILING 9/001-PHOTON. Final downlink from the outbound probe launched in September 1977 will be logged before the end of 2030, after which the Deep Space Network schedules no further passes. Verification is a receive window that stops recurring in the DSN's public station-allocation records. Signal strength at last contact: on the order of 10^{-19} watts. The Bureau classifies this as a last-photon problem of the first kind — the transmitter outlived by its audience, the silence arriving twenty-three hours after it began.

Resolution source: NASA Deep Space Network public station-allocation records (DSN Now)

SYZYGY 8::1 — DEEP TIME & FIRST PRINCIPLES

> *Geological repositories, stellar-scale energy, results still checkable in ten thousand years. The long instruments.*

BIS-F-2026-0010 — deep-borehole disposal

Zone 8 · Horizon 2035 · Swerve 4.902 · Status: pending

Filed under vertical archives. By 2035 a licensed demonstration emplaces sealed radioactive capsules in a borehole exceeding one kilometer of crystalline basement rock, retrieval demonstrated once for the regulator and never intended again. The national nuclear regulator's public licensing docket and the operator's published emplacement report constitute the record: capsule count, casing schedule, seal formulation, in triplicate. The Bureau notes that the repository's defining instrument is not the canister but the depth, and depth requires no maintenance.

Resolution source: US NRC public licensing docket (ADAMS); operator's published borehole emplacement report

BIS-F-2026-0011 — space-solar-power

Zone 8 · Horizon 2037 · Swerve 5.921 · Status: pending

By 2037 an orbital platform will have beamed no less than one megawatt-hour, cumulative, of microwave power to a licensed ground rectenna, the authorization traceable through FCC experimental filings and the ITU frequency registry, with aperture, duty cycle, and end-to-end efficiency published. The Bureau observes that the sun has transmitted on an unlicensed basis for 4.6 billion years and that the paperwork is only now catching up.

Resolution source: FCC experimental license filings; ITU frequency registry

BIS-F-2026-0012 — carbon-mineralization

Zone 8 · Horizon 2034 · Swerve 6.113 · Status: pending

A basalt-hosted storage facility will inject and verifiably mineralize more than one million tonnes of carbon dioxide within a single calendar year by 2034. Tracer assays, monitoring-well logs, and core samples are filed to the Class VI well docket and the national geological survey, where any clerk with patience may audit whether the stone did as instructed.

Resolution source: US EPA Class VI well docket; national geological survey monitoring filings

BIS-F-2026-0013 — semiconductors

Zone 1 · Horizon 2041 · Swerve 6.408 · Status: pending

Requisition: one logic family qualified for a continuous century. By 2041 a radiation-hardened semiconductor process is qualified under a published JEDEC-class standard for one hundred years of unattended operation in repository and deep-borehole monitoring service. Wear-out physics, not marketing, writes the final line of the datasheet. Verification is by the standards register and the published qualification report.

Resolution source: JEDEC-class standards register; published qualification report

BIS-F-2026-0014 — fusion ✔ manifested

Zone 8 · Horizon 2027 · Swerve 6.845 · Status: manifested

Filed without ceremony. A privately operated high-field compact tokamak reports scientific gain exceeding unity from deuterium-tritium operation no later than 2027; the shot record enters the arXiv physics.plasm-ph archive and a refereed journal. Q greater than one is a threshold in bookkeeping, not in physics — the ledger moves one column. Confirmation is by published shot number and measured neutron yield, not by press event.

Resolution source: arXiv physics.plasm-ph preprint record; refereed journal publication of shot data
Resolution note: Manifested: shot record and neutron yield published in the fourth quarter of 2027, Q = 1.09 by cross-calibrated calorimetry; the ledger moved one column and the Bureau initialed the margin.

BIS-F-2026-0015 — storage-media

Zone 1 · Horizon 2029 · Swerve 7.302 · Status: pending

A national memory institution will, before end-2029, accession a statutory record set onto femtosecond-written fused-silica media rated beyond one hundred thousand years at ambient conditions, and will publish the decoder specification as an open standard in the same accession. Resolution rests with the archive's public accession catalog and the standards body's register. The Bureau notes for the file that the deliverable of record is the reading instructions, not the glass.

Resolution source: National archive public accession catalog; open standards body register

BIS-F-2026-0016 — lunar-logistics

Zone 8 · Horizon 2030 · Swerve 7.556 · Status: pending

Before end-2030 a commercial lunar lander, manifested under a national space agency's delivery program, emplaces an etched-nickel analog archive exceeding ten terabits on the lunar surface. Landing coordinates are published; the site is imaged by an orbiting reconnaissance camera; manifest, payload registry, and post-landing imagery constitute the resolution record. Estimated shelf life of the deposit is one hundred million years. The Bureau requests the coordinates be etched on the archive itself.

Resolution source: NASA CLPS mission manifest and payload registry; Lunar Reconnaissance Orbiter Camera public image archive

BIS-F-2026-0017 — foundational-physics

Zone 1 · Horizon 2031 · Swerve 8.037 · Status: pending

The mass ordering of the neutrino sector, fixed in the universe's first second and queued in ours for roughly a century, will be determined at three-sigma confidence or better by a large liquid-scintillator observatory

before end-2031. The Bureau schedules this under routine backlog clearance. Resolution source: the arXiv hep-ex preprint record and the collaboration's refereed publication.

Resolution source: arXiv hep-ex preprint record; collaboration refereed publication

BIS-F-2026-0018 — geological-repository

Zone 8 · Horizon 2032 · Swerve 8.714 · Status: pending

Form KBS-3 is stamped in advance. By 2032 a second national deep geological repository — crystalline bedrock, copper canister, bentonite buffer — receives its first emplacement of spent fuel, an event verifiable in the Swedish Radiation Safety Authority's licensing docket and the IAEA Joint Convention national reports. Emplacement date, canister serial, and gallery coordinates are to be recorded in triplicate; the third copy is addressed to whoever digs it up.

Resolution source: Swedish Radiation Safety Authority (SSM) licensing docket; IAEA Joint Convention national reports

SYZYGY 7::2 — ORBITAL MECHANICS & DUPLICITY

> *Space logistics, debris, rendezvous, dual-use ambiguity. Filings about things that are two things at once.*

BIS-F-2026-0019 — cislunar navigation

Zone 7 · Horizon 2044 · Swerve 3.940 · Status: pending

By 2044 navigation and timing at the lunar south pole will be furnished by relay spacecraft conforming to a published interoperability specification — one signal answering to two flags — and within a single week two landers of different national origin will log position fixes from the same beacons. The fix accuracy will be better than fifty meters, stated in the mission reports. Resolution: participating agencies' public mission reports and the specification's conformance registry.

Resolution source: space agency public mission reports; LunaNet-class interoperability specification conformance registry

BIS-F-2026-0020 — quantum communications

Zone 2 · Horizon 2035 · Swerve 4.871 · Status: pending

Two national metrology institutes on different continents will, by 2035, exchange cryptographic key via satellite-distributed entanglement without trusted-node relay, at a sustained rate above one bit per second. The result will be filed as physics and cited as infrastructure. Resolution: peer-reviewed publication with its arXiv quant-ph preprint record, plus the institutes' published measurement reports.

Resolution source: peer-reviewed journal publication; arXiv quant-ph preprint record; national metrology institute reports

BIS-F-2026-0021 — rendezvous and proximity operations

Zone 7 · Horizon 2038 · Swerve 5.508 · Status: pending

Filing anticipates the first commercially licensed close inspection of a foreign-operated spacecraft — imaging inside one kilometer, by consent of both operators, under a published remote-sensing license — no later than 2038. Consent will be the novelty. The capability will not be. Resolution: the licensing agency's public license registry and the ephemeris record of the approach in the public satellite catalog.

Resolution source: commercial remote-sensing license public registry; public satellite catalog ephemeris records (space-track.org)

BIS-F-2026-0022 — debris environment × rotted

Zone 7 · Horizon 2027 · Swerve 6.442 · Status: rotted

A single fragmentation event in sun-synchronous orbit will generate more than five hundred cataloged objects and compel avoidance maneuvers by at least three unaffiliated constellation operators within seventy-two hours, before the end of 2027. The Bureau has pre-stamped the incident report; only the date field is blank. Resolution: the ESA Space Debris Office annual environment report and conjunction data message volumes published by the 18th Space Defense Squadron.

Resolution source: ESA Space Debris Office annual environment report; 18th Space Defense Squadron conjunction data message volumes
Resolution note: Rotted: no qualifying fragmentation by 31 December 2027; the incident report remains pre-stamped, its date field blank, and the file is returned to the drawer unresolved.

BIS-F-2026-0023 — LEO positioning, navigation, and timing

Zone 2 · Horizon 2041 · Swerve 7.126 · Status: pending

A low-Earth-orbit constellation operating as a commercial positioning service will be designated a complementary PNT source by a national timing authority before 2041, with signal-in-space specifications published and performance monitored in quarterly public reports. The same downlink carries a data service. Auditors of one arrive as customers of the other. Resolution: the timing authority's public designation notice and the operator's FCC filings.

Resolution source: national timing authority public designation notice; FCC filings

BIS-F-2026-0024 — optical crosslinks

Zone 2 · Horizon 2029 · Swerve 7.903 · Status: pending

An optical inter-satellite link will close at one hundred gigabits per second between terminals built by two unrelated manufacturers, on orbit, no later than 2029. Conformance will be claimed against the Space Development Agency's published optical communications terminal standard and disclosed through FCC experimental license filings. The Bureau notes for the record that interoperable light is also unmonitorable light, and closes the file.

Resolution source: FCC experimental license filings; SDA optical communications terminal standard conformance disclosures

BIS-F-2026-0025 — active debris removal ✓ manifested

Zone 7 · Horizon 2027 · Swerve 8.214 · Status: manifested

Before the end of 2027, a commercial servicing vehicle will capture a non-cooperative rocket upper stage of at least two tonnes and place it on a controlled reentry trajectory. Capture will be performed without a prepared docking fixture, at a residual tumble rate under five degrees per second. Resolution: object decay entry in the public satellite catalog maintained by the 18th Space Defense Squadron, cross-referenced against the operator's national reentry authorization.

Resolution source: 18th Space Defense Squadron public satellite catalog (space-track.org) decay records; national reentry authorization
Resolution note: Manifested: capture and controlled reentry logged in the public catalog in November 2027; residual tumble at grapple 3.8 degrees per second, within tolerance, and the decay entry filed without comment.

BIS-F-2026-0026 — GNSS authentication

Zone 7 · Horizon 2031 · Swerve 8.677 · Status: pending

By 2031 a civil aviation regulator will publish airworthiness guidance naming satellite navigation message authentication as an acceptable means of compliance for spoofing mitigation, following declaration of full operational capability for Galileo's authentication service. A receiver that can prove where a signal came from is one instrument; a receiver that can prove where it did not is another; certification will require both. Resolution: EUSPA service definition documents and the regulator's public rulemaking docket.

Resolution source: EUSPA Galileo OSNMA service definition document; civil aviation regulator public rulemaking docket

BIS-F-2026-0027 — orbital downmass logistics

Zone 2 · Horizon 2033 · Swerve 9.113 · Status: pending

By 2033 a commercial reentry capsule will complete twelve licensed cargo returns within a single calendar year, each reentry authorized under a Part 450 license appearing in the FAA commercial launch and reentry license database, recovered payload mass exceeding one hundred kilograms per flight. Downmass becomes a scheduled service rather than an event. The Bureau observes that infrastructure is what remains when the applause stops recurring.

Resolution source: FAA commercial launch and reentry license database

SYZYGY 6::3 — FUSION & TRIANGULATION

> *Plasma confinement, tritium accounting, navigation without satellites. Closing loops that have never closed.*

BIS-F-2026-0028 — pulsar navigation

Zone 3 · Horizon 2042 · Swerve 4.923 · Status: pending

A deep-space probe beyond ten astronomical units determines its own position in 2042 to within two kilometres using millisecond-pulsar timing alone, no ground station in the loop for ninety days. The galaxy has been broadcasting a timing network since before the Bureau's founding; this filing merely records the first invoice raised against it. Resolution per the operating agency's navigation reports and the arXiv astro-ph record.

Resolution source: Deep-space mission navigation reports; arXiv astro-ph preprint record

BIS-F-2026-0029 — space interferometry

Zone 3 · Horizon 2038 · Swerve 5.542 · Status: pending

Three spacecraft, 2.5 million kilometres apart at every vertex, acquire simultaneous laser phase lock across all arms in 2038 — the first closed interferometric loop ever drawn at planetary scale. Lock is held long enough to certify picometre strain sensitivity against the pre-launch requirement. The triangle, once closed, is not reopened voluntarily. Resolution per the operating space agency's mission status bulletins and the arXiv:gr-qc preprint record.

Resolution source: Space agency mission operations bulletins; arXiv:gr-qc preprint record

BIS-F-2026-0030 — terrestrial navigation × rotted

Zone 3 · Horizon 2027 · Swerve 6.115 · Status: rotted

A terrestrial low-frequency navigation broadcast, Loran-class, resumes scheduled service over a national coastline in 2027 after two decades of decommissioned silence, licensed as a complementary positioning source and receivable on a chip that costs less than its antenna. Mariners who never stopped carrying the old receivers will be briefly, quietly correct. Resolution per FCC filings and the national coast guard's Notices to Mariners.

Resolution source: FCC filings; national coast guard Notices to Mariners
Resolution note: Rotted: the appropriation lapsed before first broadcast and no carrier was detected at the appointed frequency; the towers stand unpowered, and the mariners remain quietly uncorroborated.

BIS-F-2026-0031 — quantum inertial navigation

Zone 3 · Horizon 2034 · Swerve 6.807 · Status: pending

A commercial vessel completes a scheduled transoceanic crossing in 2034 navigating solely on a cold-atom interferometric inertial unit, GNSS antennas sealed under witness tape, accumulated position error at berth under one nautical mile after eleven days. The classification society enters a new notation in its register for the occasion. Insurance follows within the fiscal year, which is the event.

Resolution source: Classification society register notations; maritime authority voyage records

BIS-F-2026-0032 — timing networks

Zone 3 · Horizon 2029 · Swerve 7.208 · Status: pending

By 2029 a continental grid operator certifies its protection-relay timing as GNSS-independent, holding phase across 4,000 kilometres of dark fiber to within 100 nanoseconds for a full calendar quarter. No satellite is consulted. The certification filing, docketed with the relevant federal energy regulator, runs to 240 pages, of which six concern physics and 234 concern liability.

Resolution source: Federal energy regulator docket (FERC eLibrary); grid reliability compliance filings

BIS-F-2026-0033 — fusion energy

Zone 6 · Horizon 2033 · Swerve 7.664 · Status: pending

In 2033 a superconducting stellarator holds a reactor-relevant plasma for sixty uninterrupted minutes, exhaust and impurity accounting stable through the full hour. No disruption occurs because, in this magnetic geometry, none is available. The control-room log for the final ten minutes records no operator input whatsoever, which the Bureau classifies as the actual result. Resolution via the device's published operational records and the peer-reviewed fusion literature.

Resolution source: Published device operational records; peer-reviewed fusion literature

BIS-F-2026-0034 — fusion energy

Zone 6 · Horizon 2031 · Swerve 8.412 · Status: pending

A privately financed tokamak will sustain a fusion gain of Q ≥ 3 for thirty consecutive seconds before the close of 2031, neutron diagnostics cross-calibrated by two national laboratories. The Bureau has pre-stamped the discrepancy report: the second laboratory's yield figure will differ by four percent, and the argument over which calorimeter to believe will outlast the plasma by eighteen months. Resolution per IAEA Fusion Energy Conference proceedings and the arXiv preprint record.

Resolution source: IAEA Fusion Energy Conference proceedings; arXiv preprint record

BIS-F-2026-0035 — tritium fuel cycle

Zone 6 · Horizon 2036 · Swerve 8.917 · Status: pending

Form T-6 (Tritium Ledger, Continuous) records its first balanced month in 2036: a breeding blanket on an operating deuterium–tritium device measures a tritium breeding ratio of 1.05 ± 0.03, bred inventory exceeding burn plus decay plus the unaccounted line. The unaccounted line does not go to zero. It never goes to zero. Resolution via peer-reviewed publication in the journal Nuclear Fusion and IAEA technical meeting records.

Resolution source: Nuclear Fusion journal publication record; IAEA technical meeting records

BIS-F-2026-0036 — metrology and timekeeping

Zone 6 · Horizon 2030 · Swerve 9.331 · Status: pending

The General Conference on Weights and Measures votes in 2030 to redefine the second against an optical transition, retiring caesium after seventy years of adequate service. The vote is unanimous, the press release is 400 words, and every clock on Earth becomes, by decree and without moving, slightly better known. Thorium-229 nuclear-clock candidates at the 10^{-19} fractional level are noted for the subsequent cycle. Resolution: the published CGPM resolution record at the BIPM.

Resolution source: BIPM published CGPM resolution record



SYZYGY 5::4 — MARKETS & MESHWORKS

> *Foundry economics, supply-chain reroutings, agent-to-agent rails as infrastructure. The machine made of institutions.*

BIS-F-2026-0037 — deep-sea engineering / subsea cables

Zone 4 · Horizon 2033 · Swerve 6.152 · Status: pending

Filing 4-0633: a transoceanic fiber cable enters service whose entire maintenance regime — fault localization, retrieval, splice, reburial — is contracted to uncrewed surface and subsea vessels, with no crewed cable ship held on standby. The maintenance authority's annual submission to the international cable protection committee records mean repair time under nine days across the first four faults, two of them in water deeper than four thousand meters.

Resolution source: International Cable Protection Committee annual reports

BIS-F-2026-0038 — optical interconnect standards ✔ manifested

Zone 5 · Horizon 2027 · Swerve 6.803 · Status: manifested

The Bureau records ratification of an implementation agreement for co-packaged optics at 3.2 terabits per second per engine, signed by fourteen members of an optical networking forum after a standards contest lasting thirty-one months. Compliance test vectors are published the same week as the agreement text. Three incompatible proprietary variants are withdrawn from sale within two quarters, their installed bases maintained under escrowed documentation.

Resolution source: OIF published implementation agreements
Resolution note: Manifested: the implementation agreement and its compliance vectors were published in the same week of 2027, as scheduled; the escrowed documentation has already been consulted twice.

BIS-F-2026-0039 — robotics / port logistics

Zone 4 · Horizon 2032 · Swerve 6.977 · Status: pending

A container terminal on the Atlantic seaboard reports, in its port authority's audited annual throughput statement, that autonomous cranes and yard vehicles executed above ninety percent of all moves for a full calendar year under a published cross-vendor interoperability profile. Seven equipment manufacturers certified against the profile; two failed and are listed as such in the registry, by serial number, without comment.

Resolution source: Port authority audited annual throughput statements; interoperability certification registry

BIS-F-2026-0040 — gigafactory logistics / energy storage

Zone 4 · Horizon 2030 · Swerve 7.291 · Status: pending

A gigafactory on the European grid discloses, through the battery-passport registry mandated under EU regulation, its ten-thousandth sodium-ion automotive pack produced on a line originally tooled for lithium iron phosphate. Changeover is documented at under ninety days, with cathode slurry, calendering, and formation stages retained unmodified. The passport entries certify three thousand cycles at C/2 and list every upstream supplier within one ocean of the plant.

Resolution source: EU battery-passport registry disclosures

BIS-F-2026-0041 — orbital logistics

Zone 5 · Horizon 2031 · Swerve 7.884 · Status: pending

During the third quarter, a cryogenic depot in low Earth orbit completes custody transfer of liquid methane between vehicles operated by two unaffiliated launch providers, under a tariff schedule filed publicly in advance. Boil-off is audited by an independent metering package whose telemetry mirrors to a public archive; the discrepancy between manifested and delivered mass is under two percent. Orbital logistics acquires its first bill of lading.

Resolution source: FCC space station filings; FAA commercial launch license database

BIS-F-2026-0042 — fusion supply chain / superconducting materials

Zone 5 · Horizon 2037 · Swerve 8.026 · Status: pending

Aggregate annual production of rare-earth barium copper oxide superconducting tape exceeds ten thousand kilometers, per capacity disclosures published by the four largest suppliers, two of which did not exist at the decade's opening. Qualification lots are cross-tested between suppliers under a shared critical-current protocol. The IAEA fusion device registry lists nine projects whose magnet procurement documents cite this interchangeable supply as a design assumption.

Resolution source: Published HTS supplier capacity disclosures; IAEA FusDIS fusion device registry

BIS-F-2026-0043 — semiconductor fabrication / foundry economics

Zone 5 · Horizon 2029 · Swerve 8.417 · Status: pending

Form 5-1129-C records the qualification, by a US foundry and a Japanese toolmaker acting as consortium, of a 28-nanometer logic process in which no lithography, deposition, or metrology step originates from a single-source vendor. Dual-sourcing at every stage was previously held to be economically impossible; the consortium's published process disclosure documents forty-one qualified second sources. The Bureau notes that the resulting line is slower, hotter, and solvent.

Resolution source: Published fab process disclosures; SEMI second-source qualification records

BIS-F-2026-0044 — agent economies as infrastructure

Zone 4 · Horizon 2041 · Swerve 8.744 · Status: pending

A national central bank's public participant register for its real-time gross settlement system admits an ancillary clearing arrangement whose members are exclusively autonomous software agents holding verifiable organizational credentials, examined under the same operational-resilience audit applied to human institutions. The first oversight report records 4.1 million settled instructions in the opening quarter, zero manual interventions, and one scheduled failover executed nineteen milliseconds early.

Resolution source: Central bank RTGS public participant register and oversight reports

BIS-F-2026-0045 — agent-to-agent payment rails

Zone 4 · Horizon 2028 · Swerve 9.212 · Status: pending

An Internet-Draft specifying machine-native payment challenges over HTTP status 402 advances to Proposed Standard, its listed authors comprising three infrastructure operators and one working group whose minutes record no unresolved objections. Interoperability events in the preceding cycle demonstrated settlement between eleven independent agent runtimes and four custodial rails, each transaction bounded by a spend mandate the payer's principal never saw. The RFC number is assigned in the ordinary course of business.

Resolution source: IETF Datatracker RFC index





Appendix — The Docket manifest

The Docket (by filing number)

| Docket № | Syzygy | Domain | Horizon | Swerve | Status | |